

Cat and Dog Image Classification

using MobileNetV2 and Flask API

Deep Learning Mini Project Report

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1. Introduction

Image classification is a foundational task in computer vision where a model learns to assign an input image to one of several predefined categories. This project develops a binary image classifier that distinguishes between cats and dogs using transfer learning with the MobileNetV2 architecture. The trained model is deployed through a Flask-based web application that lets a user upload an image and instantly view the predicted class along with the model's confidence score.

2. Objective

- Build an image classification model using the MobileNetV2 pretrained architecture.
- Train the model on the provided Cat and Dog image dataset.
- Apply appropriate image preprocessing before training.
- Evaluate the trained model using accuracy and loss metrics on a validation set.
- Develop a Flask web application that accepts an uploaded image and predicts Cat or Dog.
- Display the uploaded image, predicted class, and confidence score in the web interface.

3. Dataset

The dataset is organized into 'dataset/train' and 'dataset/validation' directories, loaded with Keras' `image_dataset_from_directory` utility, which infers class labels ('cat' and 'dog') directly from the folder structure. Labels are encoded in binary mode, since this is a two-class problem.

Property	Value
Classes	Cat, Dog
Training batches per epoch	63
Batch size	32 (Keras default)
Image size	160 x 160 x 3

4. Data Preprocessing

- All images resized to 160 x 160 pixels to match the MobileNetV2 input shape.
- Pixel values rescaled using a `Rescaling(1/127.5)` layer, mapping inputs to the `[-1, 1]` range expected by MobileNetV2.
- Images loaded in batches directly from the directory structure using `image_dataset_from_directory`, with a separate directory used for validation.

5. Model Architecture

The model uses transfer learning with MobileNetV2 as a frozen feature extractor, pretrained on ImageNet. A global average pooling layer condenses the spatial feature maps, followed by a single dense output neuron with a sigmoid activation for binary classification.

- Base model: MobileNetV2 (input shape 160x160x3, `include_top=False`, `weights='imagenet'`)
- Base model trainable: False (frozen feature extractor)
- `Rescaling(1/127.5)` input layer

- GlobalAveragePooling2D layer
- Dense(1, activation='sigmoid') output layer

6. Training Configuration

- Optimizer: Adam
- Loss function: Binary Crossentropy
- Metric: Accuracy
- Epochs: 5

7. Training Results

The table below reports the training and validation accuracy/loss recorded at the end of each epoch.

Epoch	Train Accuracy	Train Loss	Val. Accuracy	Val. Loss
1	90.25 %	0.2483	96.10 %	0.1168
2	97.20 %	0.0944	96.60 %	0.0989
3	97.90 %	0.0723	96.90 %	0.0860
4	97.95 %	0.0601	96.80 %	0.0880
5	98.45 %	0.0497	96.70 %	0.0809

Validation accuracy improved sharply from 96.10% after the first epoch to a peak of 96.90% at epoch 3, while validation loss reached its lowest point of 0.0809 at the final epoch. Training accuracy reached 98.45% by epoch 5, indicating the model fit the training data well without a widening gap between training and validation performance, suggesting minimal overfitting over the 5 training epochs.

8. Sample Predictions

Two independent test images were used to verify the trained model's predictions:

Sample Image	Predicted Class	Confidence
Web app upload (Figure 2)	Cat	99.95 %
SAMPLE2.jpg (prediction.py)	Dog	99.97 %

9. Flask Web Application

The trained model was integrated into a Flask web application. The interface allows a user to drag and drop or select an image, then click 'Predict Image' to view the uploaded photo alongside the predicted class and confidence score, visualized with a colored confidence bar.

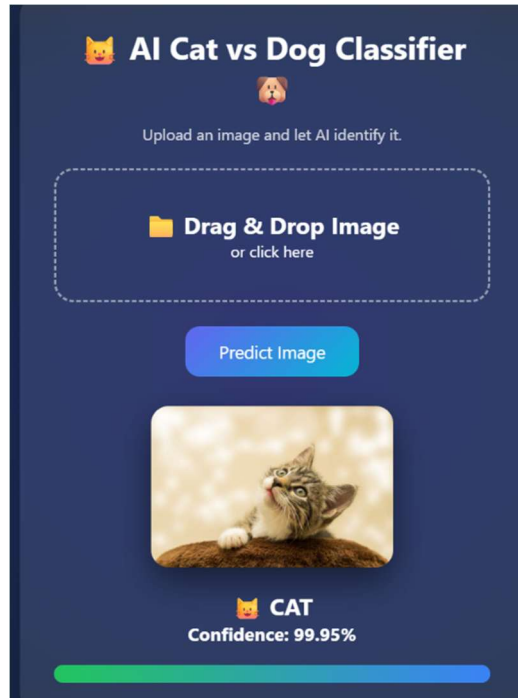


Figure 1: Cat vs Dog Classifier web interface, correctly predicting “Cat” with 99.95% confidence.

10. Conclusion

The Cat vs Dog classifier built with a frozen MobileNetV2 backbone achieved strong performance after only 5 training epochs, reaching a validation accuracy of 96.70% with low validation loss. The model generalized well on unseen sample images during manual testing, correctly identifying both a cat and a dog with confidence scores above 99.9%. The completed Flask web application successfully provides an accessible interface for real-time predictions, meeting all objectives of the assignment.

Future improvements could include fine-tuning the top layers of MobileNetV2, expanding the dataset with further augmentation, and adding batch inference or additional animal classes to extend the application's scope.